Roll No. | | | | | | | |

End Semester Back Paper Examination-2022

Name of the Program: B.Tech (CSE) — *Semester: IV*

Paper Name: Computer Organization — *Course Code: TCS-404*

Time: 3 Hour's — **Maximum Marks: 100**

Note:

(i) All questions are compulsory.
(ii) Answer any two questions among a,b,c in each question
(iii) Total marks in each question are 20 marks
(iv) Each question carries 10 marks

Q.1	**(10x2=20 Marks)**	CO-1
a)	Compare The Von Neumann and Non Von Neumann Model.	
b)	Perform signed multiplication (+12) and (-8) using Booth's Multiplication Algorithm.	
c)	Perform division of unsigned integer 12/3 using restoring division algorithm.	
Q.2	**(10x2=20 Marks)**	CO-2
a)	Differentiate between the Hardwired CU and the Micro-programmed CU?	
b)	What is instruction format? Explain instruction cycle with the help of flow chart.	
c)	What are Shift Micro operations? Starting from initial value of R= 100010110 determine the sequence of binary values in R after a Logical shift left, followed by a circular shift right, followed by an arithmetic shift Left and circular shift right..	
Q.3	**(10x2=20 Marks)**	CO-3
a)	What are Interrupts? Briefly Explain various types of interrupts.	
b)	Explain IOP Processor with the help of block diagram	
c)	Define I/O Interface. What are the requirements of it? Explain its structure	
Q.4	**(10x2=20 Marks)**	CO-4
a)	What is Memory Hierarchy; Show the memory hierarchy with the help of a diagram showing speed, cost and size.	
b)	Write short notes on: I. RAM & ROM	

	ii. EPROM iii. Cache Memory iv. Virtual memory	
c)	Explain LRU cache replacement policy with a suitable explains.	
Q.5	(10x2=20 Marks)	
a)	What is pipeline? Explain Instruction pipelining with the help of an example.	CO-5
b)	A non pipelined system takes 60 ns to process a task. The same task is processed via 5 Stage pipeline having their delays as 3 ns, 2ns, 5ns, 15ns, 3ns. The buffer registers with delay of 1 ns each are used in between the stages. What is the speedup achieved for 60 tasks?	
c)	Differentiate between RISC & CISC architecture in detail	

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